PRODUCT USER MANUAL

For Atlantic -Iberia Biscay Ireland- Wave Reanalysis Product

IBI_MULTIYEAR_WAV_005_006
Issue: 3.3

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	**23/12/2017**	all	**[EIS March 2018: IBI-V4 version]** **First release** of the IBI-MFC MY **WAV-V4** product. Temporal coverage from 01/01/1992 to 31/12/2016	M. G. Sotillo A. Amo L. Aouf C. Toledano	M. G. Sotillo
1.1	**15/01/2019**	all	**[EIS April 2019: IBI-V4 version]** **Temporal extension:** from 01/01/2017 to 30/05/2018	M. G. Sotillo A. Amo L. Aouf C. Toledano	M. G. Sotillo
1.2	**08/04/2019**	all	**[EIS July 2019: IBI-V4 version]** **Temporal extension:** from 31/05/2018 to 30/12/2018. New dataset for static files.	M. G. Sotillo A. Amo L. Aouf C. Toledano	M. G. Sotillo
1.3	**06/12/2019**		**[EIS March 2020: IBI-V4 version]** **Temporal extension:** from 31/12/2018 to 30/06/2019.	M. G. Sotillo A. Amo L. Aouf C. Toledano	M. G. Sotillo
1.4	**03/04/2020**		**[EIS July 2020: IBI-V4 version]** **Temporal extension:** from 01/07/2019 to 30/12/2019.	M. G. Sotillo A. Amo L. Aouf C. Toledano	M. G. Sotillo
2.0	**10/09/2020**		**[EIS December 2020: IBI-V6 version]** **First release** of the IBI-MFC MY **WAV-V6** product. Temporal coverage from 01/01/1993 to 31/12/2019. Update of the MFWAM model code. Horizontal resolution increased to 5 km. Implementation of SWH altimeter data assimilation using optimal interpolation algorithm and off-line coupling of current forcing.	L. G. San Martín E. Barrera M. G. Sotillo A. Amo L. Aouf C. Toledano	M. G. Sotillo
2.1	**10/09/2021**		**[EIS December 2021: IBI-V6 version]** **Temporal extension:** from 30/12/2019 to 30/12/2020	M. G. Sotillo A. Amo L. Aouf C. Toledano	M. G. Sotillo
2.2	**08/08/2022**	all	**[EIS November 2022: IBI-V6 version]** **Temporal extension:** from 2020-12-31 to 2021-12-31. Adaptation of the document to the new template.	M. G. Sotillo A. Amo L. Aouf C. Toledano	M. G. Sotillo
3.0	**16/06/2023**	all	**[EIS November 2023: IBI-V7 version]** **First release** of the IBI-MFC MY **WAV-V7** product. Temporal coverage from 01/01/1993 to 30/12/2022.	C. Toledano A. Amo L. Aouf R. Aznar	S. Cailleau M. G. Sotillo

			Update of the MFWAM model code. Horizontal resolution increased from 1/20 to 1/36°. New variables on the highest wave. New climatological dataset. Update of the static files. Renaming of the datasets. New format of the PUM document.		
3.1	**31/05/2024**	All	**[EIS November 2024: IBI-V7 version]** Extension of MY wave product back to 1980 Temporal extensions up to 2023 Delivery of air-sea fluxes for WAV MYP from 1980	B. Gómez L. Castrillo L. Aouf R. Aznar	S. Cailleau M. G. Sotillo
3.2	**28/03/2025**		**[Climatology files]** Metadata modification and addition of the new variable climatology_bounds.	B. Gómez L. Castrillo L. Aouf R. Aznar	S. Cailleau S. Ciliberti
3.3	**28/05/2025**		**[EIS November 2025: IBI-V7 version]** System upgrade to produce INTERIM.	B. Gómez L. Castrillo L. Aouf R. Aznar S. Ciliberti	S. Cailleau S. Ciliberti

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GLOSSARY AND ABBREVIATIONS

CF	Climate Forecast (convention for NetCDF)
CMEMS	Copernicus Marine Service
ECMWF	European Centre for Medium Range Weather forecast
IBI	Iberian-Biscay-Ireland
INS	In situ
NRT	Near Real Time
MDS	Marine Data Store
MFC	Monitoring and Forecasting Centre
MY	Multi Year
NetCDF	Network Common Data Format
PUM	Product User Manual
QUID	Quality Information Document
SWH	Significant Wave Height
TAC	Thematic Assembly Centre

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document is the Product User Manual (PUM) for the Atlantic Iberian-Biscay-Ireland Wave (IBI-WAV) Multi-Year (MY) product **IBI_MULTIYEAR_WAV_005_006**, which provide continuous data from **01/01/1980 until M-4**. Datasets are composed of reanalysis data as soon as available and completed with interim system files meanwhile.

This product is defined as a standard grid at **1/36 degree**, (approx. 2-3 km), extending from 19°W to 5°E and 26°N to 56°N.

The product is organized in the following datasets:

1. **cmems_mod_ibi_wav_my_0.027deg_PT1H-i:** containing hourly instantaneous fields, including wave Height, Period and Direction for total spectrum and fields of Wind Wave (or wind sea), Primary Swell Wave and Secondary Swell for partitioned wave spectra; and the highest wave variables, such as maximum crest height and maximum crest-to-trough height.
2. **cmems_mod_ibi_wav_my_0.027deg_static:** containing the coordinates, bathymetry and mask files used to generate this IBI product.
3. **cmems_mod_ibi_wav_my_aflux_0.027deg_P1H-i :** containing hourly instantaneous fields of the energy flux into ocean and the ocean side stress.
4. **cmems_mod_ibi_wav_my_0.027deg-climatology_P1M-m**: containing the climatological parameters (monthly mean) of significant wave height (VHM0) and mean period (VTM02), for the period 1993-2016, and climatology_bounds.

The IBI-MFC Production Unit (run by NOW Systems with the support of CESGA, in terms of supercomputing resources) is responsible for the generation and delivery of the product.

The product is available on-line and disseminated through the Dissemination Unit of the Copernicus Marine Service after automatic and human quality controls. Files downloaded are in **NetCDF-4** format and follow **CF-1.8** convention.

The IBI-WAV MY modelling system and validation are extensively described in the Quality Information Document, while a short quality overview of qualified Essential Ocean Variables is provided in the Synthesis Quality Overview: both documents are accessible through the dedicated IBI-WAV MY product page.

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

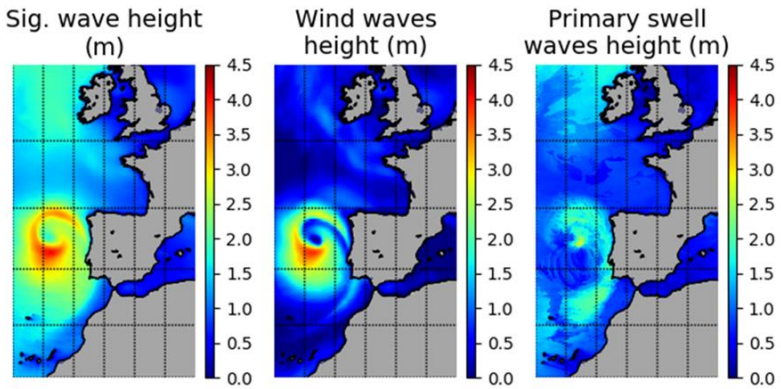
Product Lines	**IBI_MULTIYEAR_WAV_005_006**
Geographical coverage	IBI : 19°W-5°E; 26°N – 56°N
Variables	• Spectral significant wave height (Hm0) • Spectral moments (0,2) wave period (Tm02) • Spectral moments (-1,0) wave period (Tm-10) • Mean wave direction from (Mdir) • Wave principal direction at spectral peak • Wave period at spectral peak /peak period (Tp) • Maximum crest trough wave height (Hc,max) • Height of the highest crest • Spectral significant wind wave height • Spectral significant primary swell wave height • Spectral significant secondary swell wave height • Mean wind wave direction from • Mean primary swell wave direction from • Mean secondary swell wave direction from • Spectral moments (0,1) wind wave period • Spectral moments (0,1) primary swell wave period • Spectral moments (0,1) secondary swell wave period • Stokes drift U • Stokes drift V • Climatology bounds • energy flux into ocean • ocean side stress • Bathymetry • Mask • Coordinates (e1t, e2t)
Product Type	Reanalysis - Interim
Available time series	01/01/1980 to M-4 from present date
Temporal resolution	Hourly instantaneous - Monthly means
Update Frequency	Reanalysis datasets are annually reprocessed to provide temporal extension. Interim datasets are updated on monthly basis.
Target delivery time	/
Horizontal resolution	1/36 degree
Number of vertical levels	1

Format	NetCDF4

b) Production System Description

The IBI-MFC provides a high-resolution wave MY product for the Iberia-Biscay-Ireland (IBI) area starting in 01/01/1980 and being regularly extended. The model system designed and implemented by Meteo-France and NOW System – the latter is in also in charge for the products post-processing and the run of the interim processing chain, with the support of CESGA in terms of supercomputing resources.

The IBI-WAV MY model configuration is based on the MFWAM model developed by Météo-France (MF), covering the same region as the IBI Ocean Wave Analysis and Forecasting product (IBI_ANALYSISFORECAST_WAV_005_005).
The product offers hourly instantaneous fields of different wave parameters, including Wave Height, Period and Direction for total spectrum; fields of Wind Wave (or wind sea), Primary Swell Wave and Secondary Swell for partitioned wave spectra; and the highest wave variables, such as maximum crest height and maximum crest-to-trough height. Also, some air-sea fluxes such us the energy flux into ocean and the ocean side stress. Additionally, climatological parameters of significant wave height (VHM0) and zero -crossing wave period (VTM02) are delivered for the time interval 1993-2016.

Domain **Resolution and grid** **Geographic coverage**	IBI (19°W-5°E; 26°N – 56°N) 1/36° (2-3 km); regular grid; 1081 x 865 This product is regional. It is defined on a regular grid at 1/36 degree (approx. 2-3 km) that goes from 26°N to 56°N in latitude and 19°W to 5°E in longitude. Latitude and longitude step is 0.02778°. The resulting horizontal grid extends to 1081x865 grid points. Interpolated from native grid (20°W-17°E; 25°N-64.6°N). 
System Name	IBI36-WAV-MY_v7
Model version	MFWAM, based on IFS-ECWAM-47R1 with the introduction of dissipation by water-depth breaking and includes Hmax parameters from Latemar project.
Horizontal resolution	1/36°
Vertical coord.	Only surface
Spectral resolution	36 directions and 30 frequencies (start at 0.035Hz)

Atmospheric forcings	Hourly ECMWF Reanalysis ERA5 (winds)
Currents forcings	Surface currents forcing taken from the IBI ocean circulation model (IBI_MULTIYEAR_PHY_005_006)
Bounday Conditions	Wave spectra at OBC from Copernicus Marine Global Ocean Wave Reanalysis System (GLOBAL_MULTIYEAR_WAV_001_032)
Initial conditions	No
Bathymetry	ETOPO1
Assimilation scheme	Optimal interpolation of SWH (**Lionello et al., 1992**) and directional wave spectra from Envisat-SAR and CFOSAT.
Assimilated observations	-Significant wave height (SWH) from altimeters: ERS1, TOPEX/POSEIDON, ERS2, GFO, Jason 2 & 3, Envisat, Saral, Cryosat-2 and Sentinel-3A; Sentinel-3. -Spectra from Envisat-SAR and CFOSAT.

c) Dataset details

Product Lines	IBI_MULTIYEAR_WAV_005_006	
Geographical coverage	IBI	
Datasets: cmems_mod_ibi_wav_my_0.027deg_PT1H-i cmems_mod_ibi_wav_my_aflux_0.027deg_P1H-i cmems_mod_ibi_wav_my_0.027deg_static cmems_mod_ibi_wav_my_0.027deg-climatology_P1M-m		
Product Type	Reanalysis	Interim
Update frequency	Yearly	Monthly
Available time series	01/01/1980 – Interim	Reanalysis – M-4
Target delivery time	/	/
Temporal resolution	Hourly instantaneous Monthly mean	

cmems_mod_ibi_wav_my_0.027deg_PT1H-i contains instantaneous data
VHM0 [m] Spectral significant wave height (Hm0) *sea_surface_wave_significant_height*
VTM02 [s] Spectral moments (0,2) wave period (Tm02) *sea_surface_wave_mean_period_from_variance_spectral_density_second_frequency_moment*
VTM10 [s] Spectral moments (-1,0) wave period (Tm-10) *sea_surface_wave_mean_period_from_variance_spectral_density_inverse_frequency_moment*
VMDR [degree]

Mean wave direction from (Mdir) *sea_surface_wave_from_direction*
VPED [degree] Wave principal direction at spectral peak *sea_surface_wave_from_direction_at_variance_spectral_density_maximum*
VTPK [s] Wave period at spectral peak / peak period (Tp) *sea_surface_wave_period_at_variance_spectral_density_maximum*
VCMX [m] Maximum crest trough wave height (Hc,max) *sea_surface_wave_maximum_height*
VMXL [m] Height of the highest crest *sea_surface_wave_maximum_crest_height*
VHM0_WW [m] Spectral significant wind wave height *sea_surface_wind_wave_significant_height*
VHM0_SW1 [m] Spectral significant primary swell wave height *sea_surface_primary_swell_wave_significant_height*
VHM0_SW2 [m] Spectral significant secondary swell wave height *sea_surface_secondary_swell_wave_significant_height*
VMDR_WW [degree] Mean wind wave direction from *sea_surface_wind_wave_from_direction*
VMDR_SW1 [degree] Mean primary swell wave direction from *sea_surface_primary_swell_wave_from_direction*
VMDR_SW2 [degree] Mean secondary swell wave direction from *sea_surface_secondary_swell_wave_from_direction*
VTM01_WW [s] Spectral moments (0,1) wind wave period *sea_surface_wind_wave_mean_period*
VTM01_SW1 [s] Spectral moments (0,1) primary swell wave period *sea_surface_primary_swell_wave_mean_period*
VTM01_SW2 [s] Spectral moments (0,1) secondary swell wave period *sea_surface_secondary_swell_wave_mean_period*
VSDX [m s-1] Stokes drift U *sea_surface_wave_stokes_drift_x_velocity*
VSDY [m s-1] Stokes drift V *sea_surface_wave_stokes_drift_y_velocity*
cmems_mod_ibi_wav_my_aflux_0.027deg_P1H-i contains instantaneous data

TAUX [Pa] Eastward ocean side stress *surface_downward_eastward_stress_due_to_ocean_viscous_dissipation*
TAUY [Pa] Northward ocean side stress *surface_downward_northward_stress_due_to_ocean_viscous_dissipation*
TLA [W m-2] Energy flux into ocean *wave_mixing_energy_flux_into_sea_water*
cmems_mod_ibi_wav_my_0.027deg_static contains the static fields for the system
IBI-MFC_005_006_mask_bathy.nc **deptho [m]** Bathymetry *sea_floor_depth_below_geoid* **mask** [] Land-sea mask: 1 = sea; 0 = land *sea_binary_mask*
IBI-MFC_005_006_coordinates.nc **e1t [m]** Cell dimension along X axis **e2t [m]** Cell dimension along Y axis
cmems_mod_ibi_wav_my_0.027deg-climatology_P1M-m contains monthly mean data
VHM0_mean [m] Spectral significant wave height (Hm0) - Climatology monthly mean *sea_surface_wave_significant_height*
VTM02_mean [s] Spectral moments (0,2) wave period (Tm02) - Climatology monthly mean *sea_surface_wave_mean_period_from_variance_spectral_density_second_frequency_moment*
Climatology bounds [hours since time reference] climatology_bounds time interval endpoints

d) Processing information

IBI-WAV MY product processing is performed according to the following schedule:

- IBI-WAV MY **Reanalysis** timeseries are produced by NOW Systems once a year, depending on availability of upstream data (reprocessed observations, open boundary conditions (OBCs), etc).

- IBI-WAV MY **Interim** datasets are operationally produced by NOW Systems on monthly basis, starting from the last available reanalysis date and covering a period up to M-4.

e) Catalogue updating

Reanalysis data are uploaded to the MDS as soon as the production of the reanalysis timeseries in native format and its post-processing are fully completed. **Reanalysis** files will overwrite the **Interim** ones for a corresponding date as soon as it is possible to reprocess the corresponding temporal period. The **Interim** system produces **Interim** datasets monthly ensuring MY time series availability until M-4.

f) Time averaging

Dataset frequency	Time averaging	Forecast lead time
montlhy	monthly means over the calendar month (first to last day of the month centered at mid of the month)	/
hourly-instantaneous	No time average	

g) Product use and impact

The IBI-WAV MY provides reconstruction of the wave conditions in the Atlantic Iberian-Biscay-Irish region from 1980 until the present, with improved resolution with respect to available Global Ocean Wave Reanalysis. It resolves bathymetric and coastal features at an improved resolution, which is crucial for applications where local detail is a requirement. Its comprehensive set of parameters allows users to characterize the wave conditions through parameters like the average and maximum wave height, predominant directions, or the distribution of wave periods. This supports the activities described in the framework of Blue Economy, and some examples of these are:

- Design of coastal infrastructures.
- Evaluation of Wave Energy Potential.
- Support to fisheries and aquiculture.
- Navigation safety.
- Coastal erosion.
- Policy making.

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article: how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

Original file format
Filename nomenclature, when downloaded through the Get function of the Copernicus Marine Toolbox or the file browser. Based on production date and field date:
CMEMS_{fileVersion}_{region}_WAV_MY_NL_{freqFlag}_{datatype}_{validDate}_{validDate}_R{bulletinDate}_{productType}.nc
Where:

- **region:** a three-letter code for the region, IBI in this case.
- **fileVersion:** vxry, where x, y are the version and release number, respectively
- **freqFlag:** the frequency of data values in the file (01hsn = hourly snapshots, i.e.: hourly instantaneous fields).
- **datatype (optional):** it is a parameter which informs about the kind of included parameters.
- **validate:** YYYYMMDD is the valid date of the fields contained in the file.
- **bulletinDate:** RYYYYMMDD is the bulletin date, i.e.: data production date
- **productType:** is a two-letter code for the product type (RE01 for reanalysis, IN01 for interim).

Examples:
CMEMS_v7r1_IBI_WAV_MY_NL_01hsn_20200101_20200101_R20200102_RE01.nc

Other information
The NetCDF-4 format is used with short integer coding for better compression, using an offset and scale factor as follows:

$$Real_value = (Display_value * scale_factor) + add_offset$$

The **missing value** for this product is: **-32767s**
Land mask is equal to **"_FillValue"** (see variable attribute on NetCDF file).
Land values are treated as missing value.

Complementary information
An exhaustive description of the files nomenclature when downloaded through the web portal is available here : Files Nomenclature and Format

More information on product and dataset nomenclatures is available here Products and Datasets nomenclature

5) FILE SIZE

DATASET NAME	NAME OF FILE	DIMENSION [GB]
cmems_mod_ibi_wav_my_0.027deg_PT1H-I	CMEMS_v7r1_IBI_WAV_MY_NL_01hsn_20200826_20200826_R20200827_RE01.nc	0.25
cmems_mod_ibi_wav_my_aflux_0.027deg_P1H-i	CMEMS_v7r1_IBI_WAV_MY_NL_01hsn_aflux_20200826_20200826_R20200827_RE01.nc	0.04
cmems_mod_ibi_wav_my_0.027deg_static	IBI-MFC_005_006_coordinates.nc IBI-MFC_005_006_mask_bathy.nc	0.01

6) STRUCTURE OF FILES

Examples of the header of output NetCDF files are inserted in **Annex,** for each dataset.

7) REFERENCES

Lionello, P., Günther, H., & Janssen, P. A. (1992). Assimilation of altimeter data in a global third-generation wave model. Journal of Geophysical Research: Oceans, 97(C9), 14453-14474. **https://doi.org/10.1029/92JC01055**

Quality Information Document
https://documentation.marine.copernicus.eu/QUID/CMEMS-IBI-QUID-005-006.pdf

8) ANNEX

Examples of structure and header of IBI_MULTIYEAR_WAV_005_006 files downloaded through the MDS.

cmems_mod_ibi_wav_my_0.027deg_PT1H-i

```
netcdf CMEMS_v7r1_IBI_WAV_MY_NL_01hsn_20190908_20190908_R20241126_RE01 {
dimensions:
	time = 24 ;
	longitude = 865 ;
	latitude = 1081 ;
variables:
	float time(time) ;
		time:_FillValue = NaNf ;
		time:standard_name = "time" ;
		time:long_name = "time" ;
		time:axis = "T" ;
		time:units = "hours since 1950-01-01" ;
		time:calendar = "gregorian" ;
	float longitude(longitude) ;
		longitude:_FillValue = NaNf ;
		longitude:standard_name = "longitude" ;
		longitude:long_name = "Longitude" ;
		longitude:units = "degrees_east" ;
		longitude:unit_long = "Degrees East" ;
		longitude:step = "0.02777863f" ;
		longitude:axis = "X" ;
	float latitude(latitude) ;
		latitude:_FillValue = NaNf ;
		latitude:standard_name = "latitude" ;
		latitude:long_name = "Latitude" ;
		latitude:units = "degrees_north" ;
		latitude:unit_long = "Degrees North" ;
		latitude:step = "0.02777863f" ;
		latitude:axis = "Y" ;
	short VHM0(time, latitude, longitude) ;
		VHM0:_FillValue = -32767s ;
		VHM0:standard_name = "sea_surface_wave_significant_height" ;
		VHM0:long_name = "Spectral significant wave height (Hm0)" ;
		VHM0:units = "m" ;
		VHM0:unit_long = "Meters" ;
		VHM0:valid_min = 0s ;
		VHM0:valid_max = 3000s ;
		VHM0:easting = "longitude" ;
		VHM0:northing = "latitude" ;
		VHM0:longitude_min = "-19.f" ;
		VHM0:longitude_max = "5.f" ;
		VHM0:latitude_min = "26.f" ;
		VHM0:latitude_max = "56.f" ;
		VHM0:WMO_code = 100 ;
		VHM0:type_of_analysis = "spectral analysis" ;
```

```
            VHM0:add_offset = 0.f ;
            VHM0:scale_factor = 0.01f ;
      short VTM02(time, latitude, longitude) ;
            VTM02:_FillValue = -32767s ;
            VTM02:standard_name =
"sea_surface_wave_mean_period_from_variance_spectral_density_second_frequency_mome
nt" ;
            VTM02:long_name = "Spectral moments(0,2)wave period(Tm02)" ;
            VTM02:units = "s" ;
            VTM02:unit_long = "Second" ;
            VTM02:valid_min = 0s ;
            VTM02:valid_max = 3000s ;
            VTM02:easting = "longitude" ;
            VTM02:northing = "latitude" ;
            VTM02:longitude_min = "-19.f" ;
            VTM02:longitude_max = "5.f" ;
            VTM02:latitude_min = "26.f" ;
            VTM02:latitude_max = "56.f" ;
            VTM02:WMO_code = 221 ;
            VTM02:type_of_analysis = "spectral analysis" ;
            VTM02:add_offset = 0.f ;
            VTM02:scale_factor = 0.01f ;
      short VTM10(time, latitude, longitude) ;
            VTM10:_FillValue = -32767s ;
            VTM10:standard_name =
"sea_surface_wave_mean_period_from_variance_spectral_density_inverse_frequency_momen
t" ;
            VTM10:long_name = "Spectral moments(-1,0)wave period(Tm-10)" ;
            VTM10:units = "s" ;
            VTM10:unit_long = "Second" ;
            VTM10:valid_min = 0s ;
            VTM10:valid_max = 3000s ;
            VTM10:easting = "longitude" ;
            VTM10:northing = "latitude" ;
            VTM10:longitude_min = "-19.f" ;
            VTM10:longitude_max = "5.f" ;
            VTM10:latitude_min = "26.f" ;
            VTM10:latitude_max = "56.f" ;
            VTM10:WMO_code = 201 ;
            VTM10:type_of_analysis = "spectral analysis" ;
            VTM10:add_offset = 0.f ;
            VTM10:scale_factor = 0.01f ;
      short VMDR(time, latitude, longitude) ;
            VMDR:_FillValue = -32767s ;
            VMDR:standard_name = "sea_surface_wave_from_direction" ;
            VMDR:long_name = "Mean wave direction from(Mdir)" ;
            VMDR:units = "degree" ;
            VMDR:unit_long = "Degree" ;
            VMDR:valid_min = -18000s ;
            VMDR:valid_max = 18000s ;
            VMDR:easting = "longitude" ;
            VMDR:northing = "latitude" ;
            VMDR:longitude_min = "-19.f" ;
            VMDR:longitude_max = "5.f" ;
            VMDR:latitude_min = "26.f" ;
```

```
            VMDR:latitude_max = "56.f" ;
            VMDR:WMO_code = 200 ;
            VMDR:type_of_analysis = "spectral analysis" ;
            VMDR:add_offset = 180.f ;
            VMDR:scale_factor = 0.01f ;
      short VPED(time, latitude, longitude) ;
            VPED:_FillValue = -32767s ;
            VPED:standard_name =
"sea_surface_wave_from_direction_at_variance_spectral_density_maximum" ;
            VPED:long_name = "Wave principal direction at spectral peak" ;
            VPED:units = "degree" ;
            VPED:unit_long = "Degree" ;
            VPED:valid_min = -18000s ;
            VPED:valid_max = 18000s ;
            VPED:easting = "longitude" ;
            VPED:northing = "latitude" ;
            VPED:longitude_min = "-19.f" ;
            VPED:longitude_max = "5.f" ;
            VPED:latitude_min = "26.f" ;
            VPED:latitude_max = "56.f" ;
            VPED:WMO_code = 999 ;
            VPED:type_of_analysis = "spectral analysis" ;
            VPED:add_offset = 180.f ;
            VPED:scale_factor = 0.01f ;
      short VTPK(time, latitude, longitude) ;
            VTPK:_FillValue = -32767s ;
            VTPK:standard_name =
"sea_surface_wave_period_at_variance_spectral_density_maximum" ;
            VTPK:long_name = "Wave period at spectral peak/peak period(Tp)" ;
            VTPK:units = "s" ;
            VTPK:unit_long = "Second" ;
            VTPK:valid_min = 0s ;
            VTPK:valid_max = 3000s ;
            VTPK:easting = "longitude" ;
            VTPK:northing = "latitude" ;
            VTPK:longitude_min = "-19.f" ;
            VTPK:longitude_max = "5.f" ;
            VTPK:latitude_min = "26.f" ;
            VTPK:latitude_max = "56.f" ;
            VTPK:WMO_code = 204 ;
            VTPK:type_of_analysis = "spectral analysis" ;
            VTPK:add_offset = 0.f ;
            VTPK:scale_factor = 0.01f ;
      short VCMX(time, latitude, longitude) ;
            VCMX:_FillValue = -32767s ;
            VCMX:standard_name = "sea_surface_wave_maximum_height" ;
            VCMX:long_name = "Maximum crest trough wave height(Hc,max)" ;
            VCMX:units = "m" ;
            VCMX:unit_long = "Meters" ;
            VCMX:valid_min = 0s ;
            VCMX:valid_max = 3000s ;
            VCMX:easting = "longitude" ;
            VCMX:northing = "latitude" ;
            VCMX:longitude_min = "-19.f" ;
            VCMX:longitude_max = "5.f" ;
```

```
        VCMX:latitude_min = "26.f" ;
        VCMX:latitude_max = "56.f" ;
        VCMX:WMO_code = 999 ;
        VCMX:type_of_analysis = "crest" ;
        VCMX:add_offset = 0.f ;
        VCMX:scale_factor = 0.01f ;
    short VMXL(time, latitude, longitude) ;
        VMXL:_FillValue = -32767s ;
        VMXL:standard_name = "sea_surface_wave_maximum_crest_height" ;
        VMXL:long_name = "Height of the highest crest" ;
        VMXL:units = "m" ;
        VMXL:unit_long = "Meters" ;
        VMXL:valid_min = 0s ;
        VMXL:valid_max = 3000s ;
        VMXL:easting = "longitude" ;
        VMXL:northing = "latitude" ;
        VMXL:longitude_min = "-19.f" ;
        VMXL:longitude_max = "5.f" ;
        VMXL:latitude_min = "26.f" ;
        VMXL:latitude_max = "56.f" ;
        VMXL:WMO_code = 999 ;
        VMXL:type_of_analysis = "crest" ;
        VMXL:add_offset = 0.f ;
        VMXL:scale_factor = 0.01f ;
    short VHM0_WW(time, latitude, longitude) ;
        VHM0_WW:_FillValue = -32767s ;
        VHM0_WW:standard_name = "sea_surface_wind_wave_significant_height" ;
        VHM0_WW:long_name = "Spectral significant wind wave height" ;
        VHM0_WW:units = "m" ;
        VHM0_WW:unit_long = "Meters" ;
        VHM0_WW:valid_min = 0s ;
        VHM0_WW:valid_max = 3000s ;
        VHM0_WW:easting = "longitude" ;
        VHM0_WW:northing = "latitude" ;
        VHM0_WW:longitude_min = "-19.f" ;
        VHM0_WW:longitude_max = "5.f" ;
        VHM0_WW:latitude_min = "26.f" ;
        VHM0_WW:latitude_max = "56.f" ;
        VHM0_WW:WMO_code = 102 ;
        VHM0_WW:type_of_analysis = "spectral analysis" ;
        VHM0_WW:add_offset = 0.f ;
        VHM0_WW:scale_factor = 0.01f ;
    short VHM0_SW1(time, latitude, longitude) ;
        VHM0_SW1:_FillValue = -32767s ;
        VHM0_SW1:standard_name = "sea_surface_primary_swell_wave_significant_height" ;
        VHM0_SW1:long_name = "Spectral significant primary swell wave height" ;
        VHM0_SW1:units = "m" ;
        VHM0_SW1:unit_long = "Meters" ;
        VHM0_SW1:valid_min = 0s ;
        VHM0_SW1:valid_max = 3000s ;
        VHM0_SW1:easting = "longitude" ;
        VHM0_SW1:northing = "latitude" ;
        VHM0_SW1:longitude_min = "-19.f" ;
        VHM0_SW1:longitude_max = "5.f" ;
        VHM0_SW1:latitude_min = "26.f" ;
```

```
            VHM0_SW1:latitude_max = "56.f" ;
            VHM0_SW1:WMO_code = 202 ;
            VHM0_SW1:type_of_analysis = "spectral analysis" ;
            VHM0_SW1:add_offset = 0.f ;
            VHM0_SW1:scale_factor = 0.01f ;
      short VHM0_SW2(time, latitude, longitude) ;
            VHM0_SW2:_FillValue = -32767s ;
            VHM0_SW2:standard_name =
"sea_surface_secondary_swell_wave_significant_height" ;
            VHM0_SW2:long_name = "Spectral significant secondary swell wave height" ;
            VHM0_SW2:units = "m" ;
            VHM0_SW2:unit_long = "Meters" ;
            VHM0_SW2:valid_min = 0s ;
            VHM0_SW2:valid_max = 3000s ;
            VHM0_SW2:easting = "longitude" ;
            VHM0_SW2:northing = "latitude" ;
            VHM0_SW2:longitude_min = "-19.f" ;
            VHM0_SW2:longitude_max = "5.f" ;
            VHM0_SW2:latitude_min = "26.f" ;
            VHM0_SW2:latitude_max = "56.f" ;
            VHM0_SW2:WMO_code = 203 ;
            VHM0_SW2:type_of_analysis = "spectral analysis" ;
            VHM0_SW2:add_offset = 0.f ;
            VHM0_SW2:scale_factor = 0.01f ;
      short VMDR_WW(time, latitude, longitude) ;
            VMDR_WW:_FillValue = -32767s ;
            VMDR_WW:standard_name = "sea_surface_wind_wave_from_direction" ;
            VMDR_WW:long_name = "Mean wind wave direction from" ;
            VMDR_WW:units = "degree" ;
            VMDR_WW:unit_long = "Degree" ;
            VMDR_WW:valid_min = -18000s ;
            VMDR_WW:valid_max = 18000s ;
            VMDR_WW:easting = "longitude" ;
            VMDR_WW:northing = "latitude" ;
            VMDR_WW:longitude_min = "-19.f" ;
            VMDR_WW:longitude_max = "5.f" ;
            VMDR_WW:latitude_min = "26.f" ;
            VMDR_WW:latitude_max = "56.f" ;
            VMDR_WW:WMO_code = 101 ;
            VMDR_WW:type_of_analysis = "spectral analysis" ;
            VMDR_WW:add_offset = 180.f ;
            VMDR_WW:scale_factor = 0.01f ;
      short VMDR_SW1(time, latitude, longitude) ;
            VMDR_SW1:_FillValue = -32767s ;
            VMDR_SW1:standard_name = "sea_surface_primary_swell_wave_from_direction" ;
            VMDR_SW1:long_name = "Mean primary swell wave direction from" ;
            VMDR_SW1:units = "degree" ;
            VMDR_SW1:unit_long = "Degree" ;
            VMDR_SW1:valid_min = -18000s ;
            VMDR_SW1:valid_max = 18000s ;
            VMDR_SW1:easting = "longitude" ;
            VMDR_SW1:northing = "latitude" ;
            VMDR_SW1:longitude_min = "-19.f" ;
            VMDR_SW1:longitude_max = "5.f" ;
            VMDR_SW1:latitude_min = "26.f" ;
```

```
        VMDR_SW1:latitude_max = "56.f" ;
        VMDR_SW1:WMO_code = 107 ;
        VMDR_SW1:type_of_analysis = "spectral analysis" ;
        VMDR_SW1:add_offset = 180.f ;
        VMDR_SW1:scale_factor = 0.01f ;
    short VMDR_SW2(time, latitude, longitude) ;
        VMDR_SW2:_FillValue = -32767s ;
        VMDR_SW2:standard_name = "sea_surface_secondary_swell_wave_from_direction" ;
        VMDR_SW2:long_name = "Mean secondary swell wave direction from" ;
        VMDR_SW2:units = "degree" ;
        VMDR_SW2:unit_long = "Degree" ;
        VMDR_SW2:valid_min = -18000s ;
        VMDR_SW2:valid_max = 18000s ;
        VMDR_SW2:easting = "longitude" ;
        VMDR_SW2:northing = "latitude" ;
        VMDR_SW2:longitude_min = "-19.f" ;
        VMDR_SW2:longitude_max = "5.f" ;
        VMDR_SW2:latitude_min = "26.f" ;
        VMDR_SW2:latitude_max = "56.f" ;
        VMDR_SW2:WMO_code = 109 ;
        VMDR_SW2:type_of_analysis = "spectral analysis" ;
        VMDR_SW2:add_offset = 180.f ;
        VMDR_SW2:scale_factor = 0.01f ;
    short VTM01_WW(time, latitude, longitude) ;
        VTM01_WW:_FillValue = -32767s ;
        VTM01_WW:standard_name = "sea_surface_wind_wave_mean_period" ;
        VTM01_WW:long_name = "Spectral moments(0,1)wind wave period" ;
        VTM01_WW:units = "s" ;
        VTM01_WW:unit_long = "Second" ;
        VTM01_WW:valid_min = 0s ;
        VTM01_WW:valid_max = 3000s ;
        VTM01_WW:easting = "longitude" ;
        VTM01_WW:northing = "latitude" ;
        VTM01_WW:longitude_min = "-19.f" ;
        VTM01_WW:longitude_max = "5.f" ;
        VTM01_WW:latitude_min = "26.f" ;
        VTM01_WW:latitude_max = "56.f" ;
        VTM01_WW:WMO_code = 223 ;
        VTM01_WW:type_of_analysis = "spectral analysis" ;
        VTM01_WW:add_offset = 0.f ;
        VTM01_WW:scale_factor = 0.01f ;
    short VTM01_SW1(time, latitude, longitude) ;
        VTM01_SW1:_FillValue = -32767s ;
        VTM01_SW1:standard_name = "sea_surface_primary_swell_wave_mean_period" ;
        VTM01_SW1:long_name = "Spectral moments(0,1)primary swell wave period" ;
        VTM01_SW1:units = "s" ;
        VTM01_SW1:unit_long = "Second" ;
        VTM01_SW1:valid_min = 0s ;
        VTM01_SW1:valid_max = 3000s ;
        VTM01_SW1:easting = "longitude" ;
        VTM01_SW1:northing = "latitude" ;
        VTM01_SW1:longitude_min = "-19.f" ;
        VTM01_SW1:longitude_max = "5.f" ;
        VTM01_SW1:latitude_min = "26.f" ;
        VTM01_SW1:latitude_max = "56.f" ;
```

```
        VTM01_SW1:WMO_code = 226 ;
        VTM01_SW1:type_of_analysis = "spectral analysis" ;
        VTM01_SW1:add_offset = 0.f ;
        VTM01_SW1:scale_factor = 0.01f ;
    short VTM01_SW2(time, latitude, longitude) ;
        VTM01_SW2:_FillValue = -32767s ;
        VTM01_SW2:standard_name = "sea_surface_secondary_swell_wave_mean_period" ;
        VTM01_SW2:long_name = "Spectral moments(0,1)secondary swell wave period" ;
        VTM01_SW2:units = "s" ;
        VTM01_SW2:unit_long = "Second" ;
        VTM01_SW2:valid_min = 0s ;
        VTM01_SW2:valid_max = 3000s ;
        VTM01_SW2:easting = "longitude" ;
        VTM01_SW2:northing = "latitude" ;
        VTM01_SW2:longitude_min = "-19.f" ;
        VTM01_SW2:longitude_max = "5.f" ;
        VTM01_SW2:latitude_min = "26.f" ;
        VTM01_SW2:latitude_max = "56.f" ;
        VTM01_SW2:WMO_code = 227 ;
        VTM01_SW2:type_of_analysis = "spectral analysis" ;
        VTM01_SW2:add_offset = 0.f ;
        VTM01_SW2:scale_factor = 0.01f ;
    short VSDX(time, latitude, longitude) ;
        VSDX:_FillValue = -32767s ;
        VSDX:standard_name = "sea_surface_wave_stokes_drift_x_velocity" ;
        VSDX:long_name = "Stokes drift U" ;
        VSDX:units = "m s-1" ;
        VSDX:unit_long = "Meters per second" ;
        VSDX:valid_min = 0s ;
        VSDX:valid_max = 100s ;
        VSDX:easting = "longitude" ;
        VSDX:northing = "latitude" ;
        VSDX:longitude_min = "-19.f" ;
        VSDX:longitude_max = "5.f" ;
        VSDX:latitude_min = "26.f" ;
        VSDX:latitude_max = "56.f" ;
        VSDX:WMO_code = 215 ;
        VSDX:type_of_analysis = "spectral analysis" ;
        VSDX:add_offset = 0.f ;
        VSDX:scale_factor = 0.01f ;
    short VSDY(time, latitude, longitude) ;
        VSDY:_FillValue = -32767s ;
        VSDY:standard_name = "sea_surface_wave_stokes_drift_y_velocity" ;
        VSDY:long_name = "Stokes drift V" ;
        VSDY:units = "m s-1" ;
        VSDY:unit_long = "Meters per second" ;
        VSDY:valid_min = -500s ;
        VSDY:valid_max = 500s ;
        VSDY:easting = "longitude" ;
        VSDY:northing = "latitude" ;
        VSDY:longitude_min = "-19.f" ;
        VSDY:longitude_max = "5.f" ;
        VSDY:latitude_min = "26.f" ;
        VSDY:latitude_max = "56.f" ;
        VSDY:WMO_code = 216 ;
```

```
            VSDY:type_of_analysis = "spectral analysis" ;
            VSDY:add_offset = 0.f ;
            VSDY:scale_factor = 0.01f ;

// global attributes:
            :Conventions = "CF-1.8" ;
            :title = "CMEMS IBI MULTI-YEAR REANALYSIS: HOURLY WAVE PRODUCTS" ;
            :source = "MFWAM-CY47R1" ;
            :domain_name = "IBI36" ;
            :field_type = "instantaneous" ;
            :field_date = "20190908" ;
            :institution = "NOWSystems-MeteoFrance" ;
            :references = "http://marine.copernicus.eu/" ;
            :contact = "https://marine.copernicus.eu/contact" ;
            :licence = "https://marine.copernicus.eu/user-corner/service-commitments-and-
licence" ;
            :comment = "" ;
}
```

cmems_mod_ibi_wav_my_aflux_0.027deg_P1H-i

```
netcdf CMEMS_v7r1_IBI_WAV_MY_NL_01hsn_aflux_20190908_20190908_R20241126_RE01 {
dimensions:
      time = 24 ;
      latitude = 1081 ;
      longitude = 865 ;
variables:
      short TLA(time, latitude, longitude) ;
            TLA:_FillValue = -32767s ;
            TLA:standard_name = "wave_mixing_energy_flux_into_sea_water" ;
            TLA:long_name = "Energy flux into the ocean" ;
            TLA:units = "W m-2" ;
            TLA:unit_long = "Watts per square metre" ;
            TLA:valid_min = -500000 ;
            TLA:valid_max = 500000 ;
            TLA:easting = "longitude" ;
            TLA:northing = "latitude" ;
            TLA:longitude_min = "-19.f" ;
            TLA:longitude_max = "5.f" ;
            TLA:latitude_min = "26.f" ;
            TLA:latitude_max = "56.f" ;
            TLA:WMO_code = 999 ;
            TLA:type_of_analysis = "spectral analysis" ;
            TLA:add_offset = 0.f ;
            TLA:scale_factor = 0.01f ;
      float time(time) ;
            time:_FillValue = NaNf ;
            time:standard_name = "time" ;
            time:long_name = "time" ;
            time:axis = "T" ;
            time:units = "hours since 1950-01-01" ;
            time:calendar = "gregorian" ;
      float longitude(longitude) ;
            longitude:_FillValue = NaNf ;
            longitude:standard_name = "longitude" ;
            longitude:long_name = "Longitude" ;
```

```
            longitude:units = "degrees_east" ;
            longitude:unit_long = "Degrees East" ;
            longitude:step = "0.02777863f" ;
            longitude:axis = "X" ;
      float latitude(latitude) ;
            latitude:_FillValue = NaNf ;
            latitude:standard_name = "latitude" ;
            latitude:long_name = "Latitude" ;
            latitude:units = "degrees_north" ;
            latitude:unit_long = "Degrees North" ;
            latitude:step = "0.02777863f" ;
            latitude:axis = "Y" ;
      short TAUX(time, latitude, longitude) ;
            TAUX:_FillValue = -32767s ;
            TAUX:standard_name =
"surface_downward_eastward_stress_due_to_ocean_viscous_dissipation" ;
            TAUX:long_name = "Eastward ocean side stress" ;
            TAUX:units = "Pa" ;
            TAUX:unit_long = "Pascals" ;
            TAUX:valid_min = -100000 ;
            TAUX:valid_max = 100000 ;
            TAUX:easting = "longitude" ;
            TAUX:northing = "latitude" ;
            TAUX:longitude_min = "-19.f" ;
            TAUX:longitude_max = "5.f" ;
            TAUX:latitude_min = "26.f" ;
            TAUX:latitude_max = "56.f" ;
            TAUX:WMO_code = 999 ;
            TAUX:type_of_analysis = "spectral analysis" ;
            TAUX:add_offset = 0.f ;
            TAUX:scale_factor = 0.01f ;
      short TAUY(time, latitude, longitude) ;
            TAUY:_FillValue = -32767s ;
            TAUY:standard_name =
"surface_downward_northward_stress_due_to_ocean_viscous_dissipation" ;
            TAUY:long_name = "Northward ocean side stress" ;
            TAUY:units = "Pa" ;
            TAUY:unit_long = "Pascals" ;
            TAUY:valid_min = -100000 ;
            TAUY:valid_max = 100000 ;
            TAUY:easting = "longitude" ;
            TAUY:northing = "latitude" ;
            TAUY:longitude_min = "-19.f" ;
            TAUY:longitude_max = "5.f" ;
            TAUY:latitude_min = "26.f" ;
            TAUY:latitude_max = "56.f" ;
            TAUY:WMO_code = 999 ;
            TAUY:type_of_analysis = "spectral analysis" ;
            TAUY:add_offset = 0.f ;
            TAUY:scale_factor = 0.01f ;

// global attributes:
            :Conventions = "CF-1.8" ;
            :title = "CMEMS IBI MULTI-YEAR REANALYSIS: HOURLY WAVE PRODUCTS" ;
            :source = "MFWAM-CY47R1" ;
```

```
            :domain_name = "IBI36" ;
            :field_type = "instantaneous" ;
            :field_date = "20190908" ;
            :institution = "NOWSystems-MeteoFrance" ;
            :references = "http://marine.copernicus.eu/" ;
            :contact = "https://marine.copernicus.eu/contact" ;
            :licence = "https://marine.copernicus.eu/user-corner/service-commitments-and-
licence" ;
            :comment = "" ;
}
```

cmems_mod_ibi_wav_my_0.027deg_static
This dataset only contains 2 files:
IBI-MFC_005_006_mask_bathy.nc
IBI-MFC_005_006_coordinates.nc

```
netcdf IBI-MFC_005_006_mask_bathy {
dimensions:
      latitude = 1081 ;
      longitude = 865 ;
variables:
      float latitude(latitude) ;
            latitude:standard_name = "latitude" ;
            latitude:units = "degrees_north" ;
            latitude:axis = "Y" ;
            latitude:step = "0.02777863f" ;
      float longitude(longitude) ;
            longitude:standard_name = "longitude" ;
            longitude:units = "degrees_east" ;
            longitude:axis = "X" ;
            longitude:step = "0.02777863f" ;
      float mask(latitude, longitude) ;
            mask:standard_name = "sea_binary_mask" ;
            mask:long_name = "Land-sea mask: 1 = sea ; 0 = land" ;
            mask:units = "1" ;
      float deptho(latitude, longitude) ;
            deptho:standard_name = "sea_floor_depth_below_geoid" ;
            deptho:long_name = "Bathymetry" ;
            deptho:units = "m" ;
            deptho:_FillValue = -32767.f ;

// global attributes:
            :Conventions = "CF-1.8" ;
            :title = "Static files for product IBI_MULTIYEAR_WAV_005_006" ;
            :domain_name = "IBI36" ;
            :institution = "NOWSystems-MeteoFrance" ;
            :references = "http://marine.copernicus.eu" ;
            :contact = "https://marine.copernicus.eu/contact" ;
            :licence = "http://marine.copernicus.eu/services-portfolio/service-commitments-and-
licence" ;
            :credit = "E.U. Copernicus Marine Service Information" ;
            :comment = "" ;
}

netcdf IBI-MFC_005_006_coordinates {
```

```
dimensions:
        latitude = 1081 ;
        longitude = 865 ;
variables:
        float latitude(latitude) ;
                latitude:standard_name = "latitude" ;
                latitude:units = "degrees_north" ;
                latitude:axis = "Y" ;
                latitude:step = "0.02777863f" ;
        float longitude(longitude) ;
                longitude:standard_name = "longitude" ;
                longitude:units = "degrees_east" ;
                longitude:axis = "X" ;
                longitude:step = "0.02777863f" ;
        float e1t(latitude, longitude) ;
                e1t:long_name = "Cell dimension along X axis" ;
                e1t:units = "m" ;
                e1t:_FillValue = -32767.f ;
        float e2t(latitude, longitude) ;
                e2t:long_name = "Cell dimension along Y axis" ;
                e2t:units = "m" ;
                e2t:_FillValue = -32767.f ;

// global attributes:
                :Conventions = "CF-1.8" ;
                :title = "Static files for product IBI_MULTIYEAR_WAV_005_006" ;
                :domain_name = "IBI36" ;
                :institution = "NOWSystems-MeteoFrance" ;
                :references = "http://marine.copernicus.eu" ;
                :contact = "https://marine.copernicus.eu/contact" ;
                :licence = "http://marine.copernicus.eu/services-portfolio/service-commitments-and-
licence" ;
                :credit = "E.U. Copernicus Marine Service Information" ;
                :comment = "" ;
}
```

```
cmems_mod_ibi_wav_my_0.027deg-climatology_P1M-m
This dataset contains 12 files:
CMEMS_v7r1_IBI_WAV_MYCLIM_1993_2016_01.nc
CMEMS_v7r1_IBI_WAV_MYCLIM_1993_2016_02.nc
CMEMS_v7r1_IBI_WAV_MYCLIM_1993_2016_03.nc
CMEMS_v7r1_IBI_WAV_MYCLIM_1993_2016_04.nc
CMEMS_v7r1_IBI_WAV_MYCLIM_1993_2016_05.nc
CMEMS_v7r1_IBI_WAV_MYCLIM_1993_2016_06.nc
CMEMS_v7r1_IBI_WAV_MYCLIM_1993_2016_07.nc
CMEMS_v7r1_IBI_WAV_MYCLIM_1993_2016_08.nc
CMEMS_v7r1_IBI_WAV_MYCLIM_1993_2016_09.nc
CMEMS_v7r1_IBI_WAV_MYCLIM_1993_2016_10.nc
CMEMS_v7r1_IBI_WAV_MYCLIM_1993_2016_11.nc
CMEMS_v7r1_IBI_WAV_MYCLIM_1993_2016_12.nc
```

```
etcdf CMEMS_v7r1_IBI_WAV_MYCLIM_1993_2016_01 {
dimensions:
```

```
    time = 1 ;
    longitude = 865 ;
    latitude = 1081 ;
    nbnd = 2 ;
variables:
    float time(time) ;
        time:standard_name = "time" ;
        time:long_name = "time" ;
        time:units = "hours since 1950-01-01 00:00:00" ;
        time:calendar = "gregorian" ;
        time:axis = "T" ;
        time:climatology = "climatology_bounds" ;
    float longitude(longitude) ;
        longitude:standard_name = "longitude" ;
        longitude:long_name = "Longitude" ;
        longitude:units = "degrees_east" ;
        longitude:unit_long = "Degrees East" ;
        longitude:step = "0.02777863f" ;
        longitude:axis = "X" ;
    float latitude(latitude) ;
        latitude:standard_name = "latitude" ;
        latitude:long_name = "Latitude" ;
        latitude:units = "degrees_north" ;
        latitude:unit_long = "Degrees North" ;
        latitude:step = "0.02777863f" ;
        latitude:axis = "Y" ;
    short VHM0_mean(time, latitude, longitude) ;
        VHM0_mean:add_offset = 0.f ;
        VHM0_mean:scale_factor = 0.01f ;
        VHM0_mean:standard_name = "sea_surface_wave_significant_height" ;
        VHM0_mean:long_name = "Spectral significant wave height (Hm0) - Climatology monthly
mean" ;
        VHM0_mean:units = "m" ;
        VHM0_mean:unit_long = "Meters" ;
        VHM0_mean:valid_min = 0s ;
        VHM0_mean:valid_max = 3000s ;
        VHM0_mean:easting = "longitude" ;
        VHM0_mean:northing = "latitude" ;
        VHM0_mean:longitude_min = "-19.f" ;
        VHM0_mean:longitude_max = "5.f" ;
        VHM0_mean:latitude_min = "26.f" ;
        VHM0_mean:latitude_max = "56.f" ;
        VHM0_mean:_FillValue = -32767s ;
        VHM0_mean:WMO_code = 100 ;
        VHM0_mean:type_of_analysis = "spectral analysis" ;
        VHM0_mean:cell_methods = "time: mean over years" ;
    short VTM02_mean(time, latitude, longitude) ;
        VTM02_mean:add_offset = 0.f ;
        VTM02_mean:scale_factor = 0.01f ;
        VTM02_mean:standard_name =
"sea_surface_wave_mean_period_from_variance_spectral_density_second_frequency_mome
nt" ;
        VTM02_mean:long_name = "Spectral moments(0,2)wave period(Tm02) - Climatology
monthly mean" ;
        VTM02_mean:units = "s" ;
```

```
        VTM02_mean:unit_long = "Second" ;
        VTM02_mean:valid_min = 0s ;
        VTM02_mean:valid_max = 3000s ;
        VTM02_mean:easting = "longitude" ;
        VTM02_mean:northing = "latitude" ;
        VTM02_mean:longitude_min = "-19.f" ;
        VTM02_mean:longitude_max = "5.f" ;
        VTM02_mean:latitude_min = "26.f" ;
        VTM02_mean:latitude_max = "56.f" ;
        VTM02_mean:_FillValue = -32767s ;
        VTM02_mean:WMO_code = 221 ;
        VTM02_mean:type_of_analysis = "spectral analysis" ;
        VTM02_mean:cell_methods = "time: mean over years" ;
    double climatology_bounds(time, nbnd) ;
        climatology_bounds:long_name = "time interval endpoints" ;

// global attributes:
        :Conventions = "CF-1.8" ;
        :title = "Monthly climatology reference period 1993-2016" ;
        :source = "MFWAM-CY47R1" ;
        :domain_name = "IBI36" ;
        :institution = "NOWSystems-MeteoFrance" ;
        :references = "http://marine.copernicus.eu/" ;
        :contact = "https://marine.copernicus.eu/contact" ;
        :licence = "https://marine.copernicus.eu/user-corner/service-commitments-and-licence" ;
        :credit = "E.U. Copernicus Marine Service Information" ;
        :comment = "Climatologies derived from cmems_mod_ibi_wav_my_0.027deg_PT1H-i" ;
        :field_type = “monthly_climatology” ;
```